

ZRF SCIENCE FAIR 2026

ভবিষ্যৎ বিজ্ঞানীর খোঁজে ২০২৬

Bangla Publishing Studio

A free, zero-setup, web-based WYSIWYG book publishing platform that takes Bengali writers from first sentence to press-ready book — including the traditional Farma printing format of Bangladeshi presses.

গবেষণা ♦ উদ্ভাবন ♦ প্রযুক্তি উন্নয়ন — Research · Innovation · Technology Development

PARTICIPANT	Topu Biswas
CLASS / YEAR	B.Sc. (Engg.) 2nd Year, Department of ICE, University of Rajshahi
GROUP	Group C (18 years or above)
SCIENTIFIC FIELD	Engineering & Innovation
PROJECT TYPE	Individual Project · Working software prototype (open-source)

SECTION 01

Project at a Glance এক নজরে প্রজেক্ট

PROJECT TITLE

Bangla Publishing Studio

SCIENTIFIC FIELD

Engineering & Innovation

GROUP / CATEGORY

Group C (18 years or above)

PROJECT TYPE

Individual

PARTICIPANT

Topu Biswas

INSTITUTION

University of Rajshahi, Rajshahi Division

STATUS

Working prototype, live & open-source

KEYWORDS

Bengali publishing · WYSIWYG · Farma imposition · PWA

SECTION 02

Project Abstract সারসংক্ষেপ

Bangla Publishing Studio is a free, web-based WYSIWYG (What-You-See-Is-What-You-Get) word processor and book designer built specifically for Bengali-language publishing. While global tools such as MS Word or Adobe InDesign treat Bengali books as an afterthought, professional Bengali typesetting still depends on costly desktop software and skilled operators that most teachers, rural writers and small publishers cannot afford. This project removes that barrier completely: a writer opens a browser, starts writing on true print-sized pages (A4, Letter, Legal, Demy, Crown and other industry-standard sizes), and produces press-ready output without installing anything or paying anything.

The platform implements a real pagination engine, page headers and footers, images with text wrapping, tables, icons, shapes, MCQ/exam boxes for coaching-centre textbooks, and Bengali fonts (Kalpurush, SolaimanLipi, Siyam Rupali, Hind Siliguri) with conjunct-safe typography. Its most distinctive capability is **Farma export** — the signature-imposition print format used by Bangladeshi printing presses (ছাপাখানা) — which rearranges pages into correct printing signatures so that any local press can directly plate and bind the book. Documents auto-save offline in the browser (IndexedDB) and the app works as an installable PWA even on low-end devices with unreliable internet.

Built with Next.js, TypeScript and TipTap, the working prototype is live and open-source, demonstrating the complete pipeline end-to-end: **write** → **design** → **paginate** → **export** print-ready PDF, DOCX, EPUB, or Farma imposition.

SECTION 03

Objectives উদ্দেশ্যসমূহ

1

Democratise Bengali book publishing

Provide a 100% free, zero-setup, browser-based alternative to expensive desktop publishing (DTP) software for teachers, students, rural writers and small publishers of Bangladesh.

2 True WYSIWYG pagination on physical paper

Implement real pagination on industry-standard paper sizes (A4, Letter, Legal, Demy, Crown) with margins and gutter, so that what the writer sees on screen is exactly what the printing press produces.

3 Digitise the traditional press workflow (Farma)

Bring the signature-imposition (ফরমা) process of Bangladeshi printing presses inside a simple one-click export, including binding-safe margins and correct page ordering.

4 Protect and promote Bengali typography

Ensure correct Unicode Bengali rendering, যুক্তাক্ষর (conjunct)-safe fonts, Bengali-first menus and typography controls so non-technical users feel at home.

5 Uninterrupted writing anywhere

Guarantee offline-first operation through a PWA with automatic local backup (IndexedDB) and one-click JSON backup/restore for regions with weak internet connectivity.

6 Open foundation for other languages

Create an open-source architecture that can be extended to other low-resource languages of Bangladesh and the region.

SECTION 04

Innovation / Novelty উদ্ভাবন

First in-browser Farma imposition exporter

Reproduces the signature page-ordering used by Bangladeshi printing presses — a workflow previously available only through professional DTP operators — directly inside a web page.

Real pagination engine in the browser

Renders true physical pages (A4 to Crown/Demy with margins and gutter) while the writer types — not a print-preview guess — eliminating the screen-versus-print mismatch of ordinary word processors.

Bengali-first user experience

Full Bengali interface, Bengali fonts, conjunct-safe spacing and letter-gap tools, MCQ/exam boxes and coaching-friendly templates designed from the real needs of Bangladeshi teachers.

Zero-setup PWA architecture

Installs like an app, auto-saves offline, runs on low-cost hardware, needs no account or server — the writer's knowledge stays on the user's own device.

One integrated pipeline

Write, design, paginate and export to Farma / print-ready PDF / DOCX / EPUB / HTML in a single tool, replacing three to four separate expensive software products.

Free and open-source

No licence fee, no watermark, no subscription — the complete source code is publicly available for verification and community improvement.

Expected Impact সম্ভাব্য প্রভাব

E

Educational impact

Thousands of teachers, tutors and coaching centres can produce their own professional textbooks, question banks and notes at near-zero cost, improving the quality and availability of Bengali learning materials nationwide.

S

Social and cultural impact

Removes the language barrier in technology: a rural writer with only a low-end phone or laptop and an occasional internet connection can write and publish a complete book in Bangla — strengthening Bengali content in the digital age.

\$

Economic impact

Cuts pre-press and typesetting costs for small publishers and self-publishing writers; the project has a clear path to a sustainable startup serving Bangladesh's book market.

T

Technical impact

Demonstrates that complex print-industry workflows can be replicated with open web technology, and the architecture can be extended to other Indic languages such as Assamese and Hindi.

G

Efficiency and environmental impact

A digital-first workflow reduces trial prints and paper waste during book production, and encourages systematic digital archiving of manuscripts.

SECTION 06

How It Works কীভাবে কাজ করে

লিখুন

WRITE

→

ডিজাইন

DESIGN

→

পেজিনেট

PAGINATE

→

ফরমা

FARMA
IMPOSITION

→

প্রিন্ট-রেডি বই

PRESS-READY
OUTPUT

LAYER	TECHNOLOGY	PURPOSE
Frontend	Next.js 16, React 19, TypeScript, Tailwind CSS	Fast, reliable, modern interface that runs on any device
Editor engine	TipTap / ProseMirror with custom Bengali extensions	WYSIWYG editing, icons, text boxes, image wrapping, MCQ boxes
Pagination engine	Custom layout engine with real page metrics	True physical pages, headers/footers, content flow between pages
Print pipeline	Dynamic CSS print rules + Farma imposition	Press-ready PDF with signature page ordering and accurate margins
Storage	IndexedDB (Dexie) + JSON backup, PWA service worker	Offline autosave, instant restore, installable app experience
Hosting	Vercel + GitHub (open-source)	Globally accessible, zero-maintenance delivery

SECTION 07

Working Prototype কার্যকর মডেল

The project is not a concept on paper — it is a **working software model** that runs today in any modern browser. The following screenshots are taken directly from the live prototype.

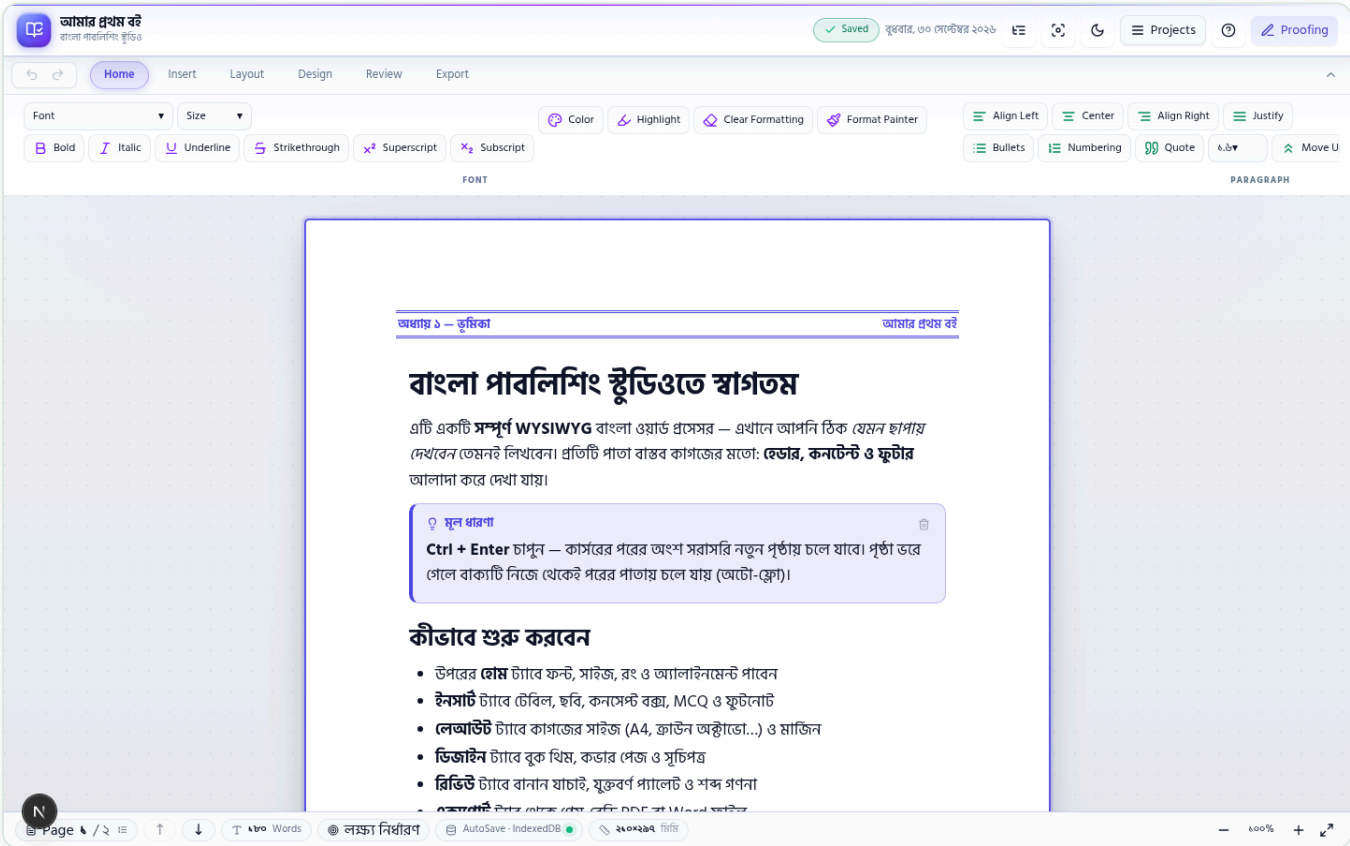


Fig. 1 — WYSIWYG Bengali editor: ribbon toolbar with font, colour, highlight, alignment, lists, quote and move tools; live page view with header; autosave status and word count.

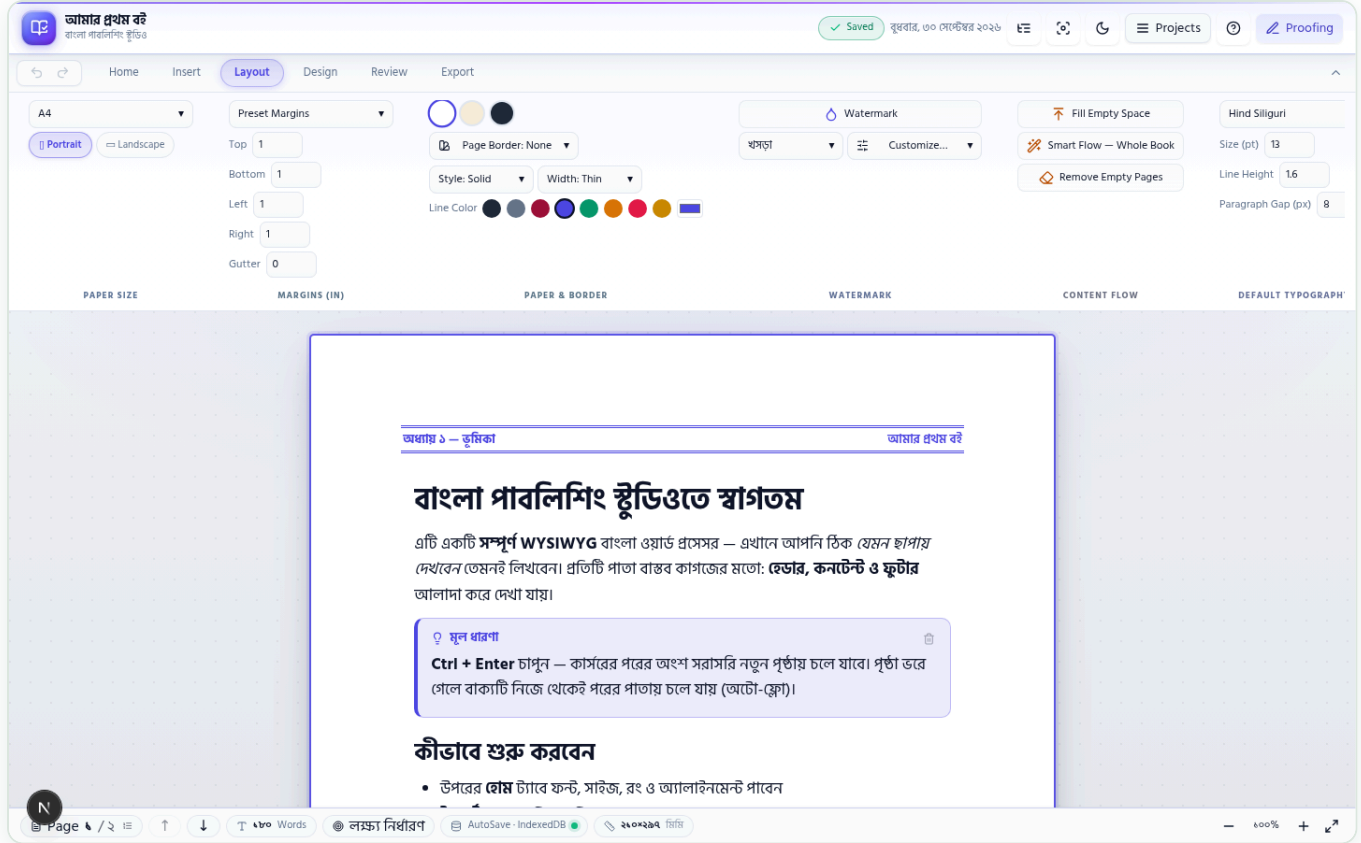


Fig. 2 — Print-layout controls: physical paper sizes (A4, Letter, Legal, Demy, Crown...), portrait/landscape, margins and gutter, page border, watermark, content flow and default typography.

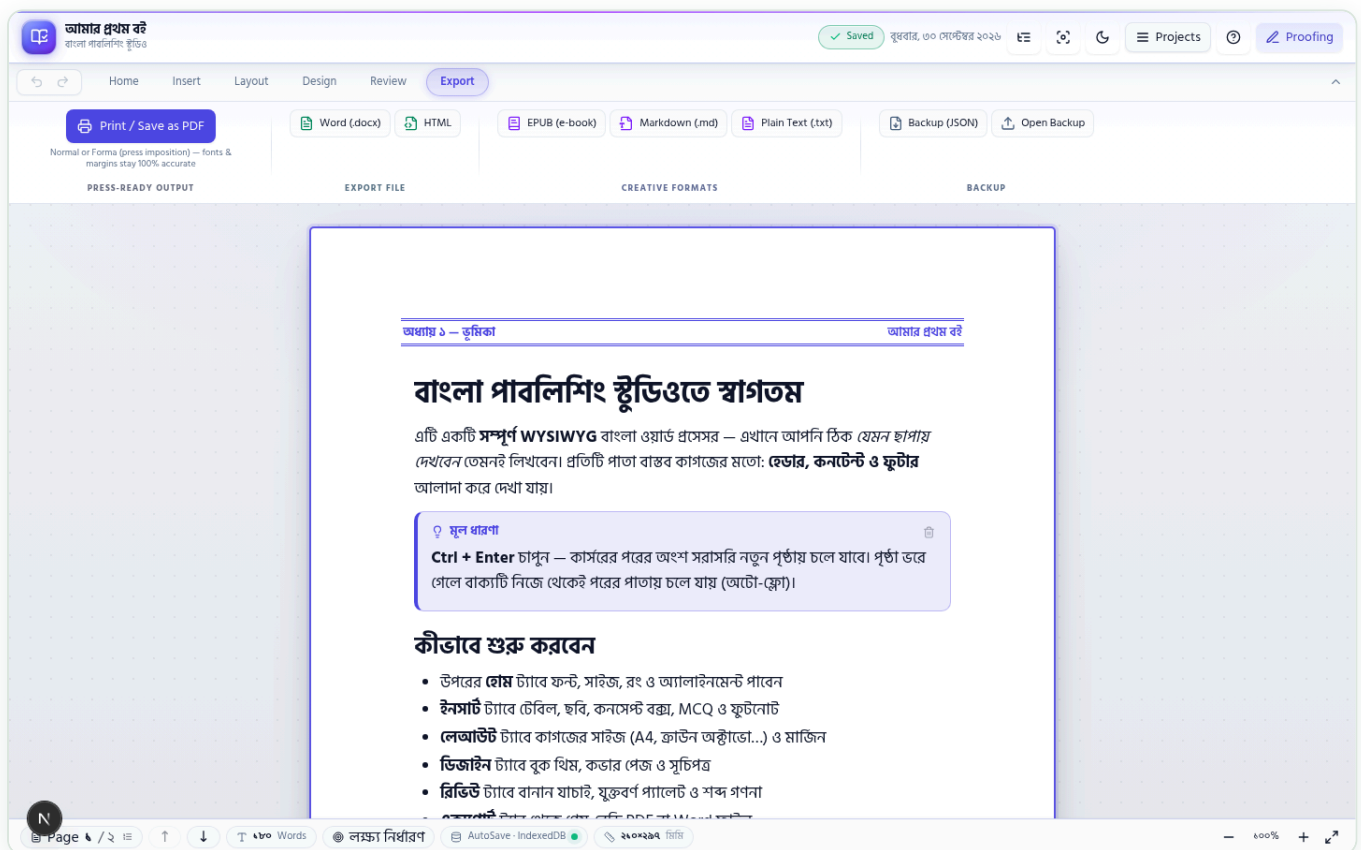


Fig. 3 — Export centre: Print / Save as PDF (normal or Farma press imposition with 100% accurate margins), Word (DOCX), HTML, EPUB e-book, Markdown, plain text and JSON backup.

Alignment with ZRF Science Fair 2026 প্রতিযোগিতার শর্ত অনুসারে

✓ Completely original and unique idea

The platform is designed and built from scratch by the participant; source code is publicly available at the GitHub repository for verification. It has not been presented in any previous competition.

✓ Written explanation with images

This Project Summary PDF itself fulfils the required "written explanation with images" — full text plus live screenshots of the working model.

✓ Working model for in-person presentation

The software prototype is the model: it can be demonstrated live, on stage, in front of the judges — writing, designing and exporting a real book in minutes.

✓ Video / PowerPoint presentation capability

A presentation (video or slide form, within 15 minutes) demonstrating the complete workflow is prepared and will be submitted/presented as required by the organisers.

✓ Individual participation (Group C)

The participant is 18 years or above and competes individually; guidance from the Department of ICE, University of Rajshahi is gratefully acknowledged.

SECTION 09

Future Roadmap ভবিষ্যৎ পরিকল্পনা

Phase 1 — Core platform COMPLETED

WYSIWYG Bengali editor, true pagination, paper sizes, headers/footers, images, tables, MCQ boxes, export to PDF/DOCX/EPUB/HTML, offline autosave and PWA installation.

Phase 2 — Press-grade hardening IN PROGRESS

Per-page header/footer override, hyperlink preservation in PDF, advanced typography controls (word/letter spacing), cross-page content reflow and production-grade Farma imposition testing with local presses.

Phase 3 — Intelligent assistance PLANNED

Bengali spelling and grammar checking, AI-assisted proofreading, automatic table of contents and index generation for long books.

Phase 4 — Ecosystem PLANNED

Template marketplace, collaborative editing, and extension to other low-resource languages of the region.

SECTION 10

Conclusion উপসংহার

Every year, countless Bengali writers complete their manuscripts but never see them become books — professional typesetting is simply out of reach. Bangla Publishing Studio crosses that final mile: true WYSIWYG pagination, Bengali-first typography and the traditional Farma press workflow combined in one free platform that converts a writer's effort directly into a printed book.

DECLARATION

Declaration of Originality অঙ্গীকারনামা

I hereby declare that the project "**Bangla Publishing Studio**" is my own original work, designed and developed by me. The idea is completely original and unique, and the project has not been submitted or presented in any previous competition or fair. The source code is openly available for verification. I agree to present my own model in person at the ZRF Science Fair 2026 and to abide by all rules and decisions of the organisers, Ziaur Rahman Foundation.

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University of Rajshahi · Group C

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30 September 2026

Source code

<https://github.com/topbook12/bangla-publishing-studio>

Submitted to

ZRF Science Fair 2026 — "ভবিষ্যৎ বিজ্ঞানীর খোঁজে ২০২৬", Ziaur Rahman Foundation